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SECTION 02

Project 2

Lecturer: Dr. Rozilawati Binti Dollah @ Md. Zain

Student Name	Matric No.
LAM YOKE YU	A23CS0233
TAN YI YA	A23CS0187
TAN ZHI MING	A23CS0189

Introduction

This report documents the process and findings of analysing the “Spotify Tracks Dataset” from Kaggle. The dataset stores Spotify songs with 114 genres and their audio features. 2 different clustering algorithms, namely the K-Means and Gaussian Mixture Model (GMM) Algorithm are applied to several specific business insights.

In terms of implementation, we decided to use Notebook instead of Data Mining tools. Due to the limitation of our knowledge, Data Mining tools such as KNIME are not as intuitive. On the other hand, notebooks give us more control over the process, allowing us to clearly see what is going on in each step.

Data Preparation

After loading the dataset, we checked if there are empty values in the dataset.

```
df = pd.read_csv('dataset.csv')
df.isnull().sum()
```

In the data preparation phase, all non-numeric and irrelevant columns — including *track_name*, *track_id*, *artists*, and *track_genre* — were removed, as they do not contribute meaningfully to clustering based on audio characteristics. Next, rows with missing values were dropped to ensure model stability and avoid preprocessing errors. The dataset was then filtered to retain only numeric features, which are suitable for mathematical modeling and distance-based algorithms. Finally, feature values were standardized using `StandardScaler` to normalize the scale of all features. This ensures that attributes like energy, valence, and tempo contribute equally to the clustering process, preventing features with larger numerical ranges from dominating the analysis.

```
drop_cols = ['track_name', 'track_id', 'artists', 'track_genre']
df_clean = df.drop(columns=drop_cols, errors='ignore')
df_clean = df_clean.dropna()
df_clean = df_clean.select_dtypes(include=[np.number])

# Standardize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df_clean)
```

Model Development

For each business insight, we need to identify its features and the targeted clustering objectives. Since both K-Means and GMM require the number of clusters as input, a relative evaluation is conducted for each business insight to determine the optimal number of clusters. Once the optimal number of clusters is identified, the algorithms are implemented.

Next, Principal Component Analysis (PCA) is adopted to reduce the dimension of the data set to two dimensions, making the clustering results more intuitive. PCA finds the directions that capture the most changes in the data and uses them to create new features called principal components. The first two components (PC1 and PC2) are used to plot data in 2D, which helps to show the effect of clustering algorithms.

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
pca_df = pd.DataFrame(X_pca, columns=["PC1", "PC2"])
```

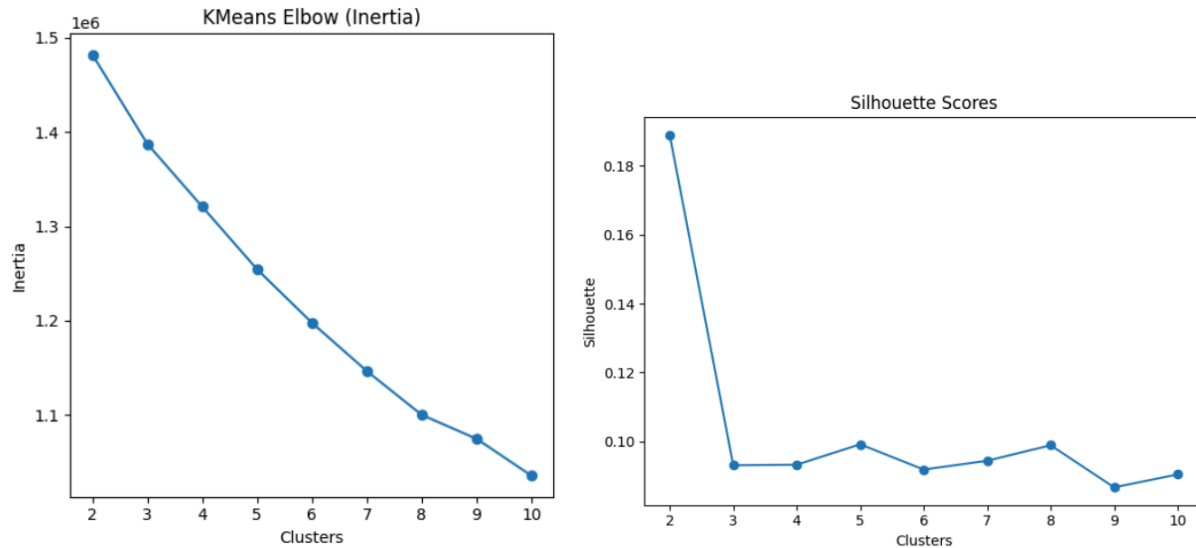
Finally, the cluster profiles are visualized using radar graphs and heatmaps or pairwise feature plots to identify the characteristics of each cluster and assign appropriate labels.

Business Insight 1: Clustering tracks based on mood

The *implicit target variable* in this insight is the **mood category** of each song, such as “Chill”, “Party”, or “Workout”. Although the dataset does not contain predefined mood labels, the aim of this project is to uncover these hidden mood-based groupings using unsupervised clustering. The mood categories represent underlying emotional or energetic states associated with each track, inferred from its audio features.

K-Means Clustering

To determine the optimal number of clusters, the elbow curve and the silhouette scores are computed.



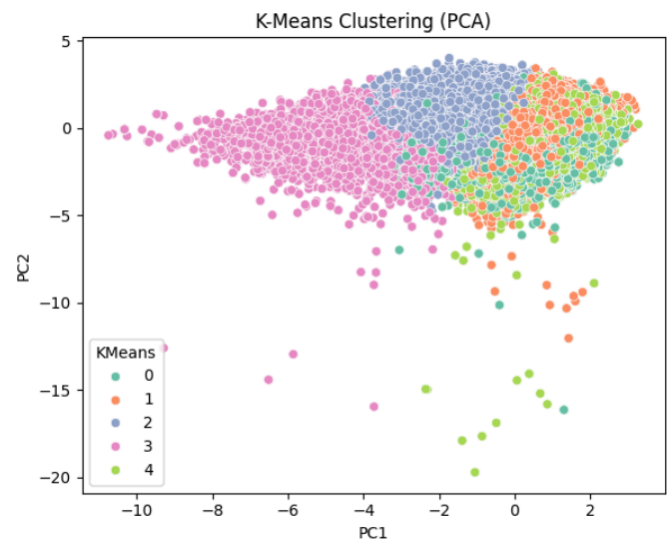
Based on this elbow-shaped curve, we can observe that there is a steep slope between 2 and 4, indicating a significant reduction in inertia. However, starting from 4, the slope begins to decrease, and the curve becomes more gradual after 5. This suggests that after this point, the improvement in compactness gradually weakens. Although the silhouette score is highest when the number of clusters is 2, clustering the data into only two groups may not provide sufficient meaningful distinctions. Instead, choosing 5 as the optimal number of clusters provides a reasonable balance - its silhouette score is slightly higher than its adjacent values, and it corresponds to the point where the rate of inertia reduction slows down. The slope of the elbow-shaped curve between 4 and 5 is still quite obvious, but after 5, the improvement becomes negligible.

K-means is applied with 5 clusters.

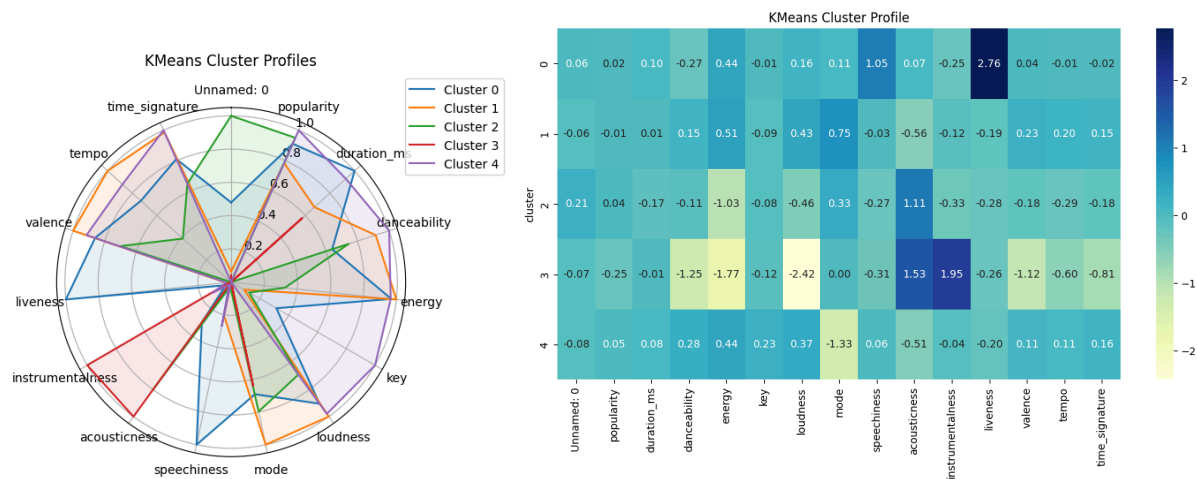
```
optimal_k = 5
```

```
kmeans = KMeans(n_clusters=optimal_k, random_state=42)
kmeans_labels = kmeans.fit_predict(X_scaled)
```

The results are visualised using PCA visualisation that reduces the data to 2D.



For more intuitive visualisation, each cluster profile is calculated and visualised.



K-Means Interpretation

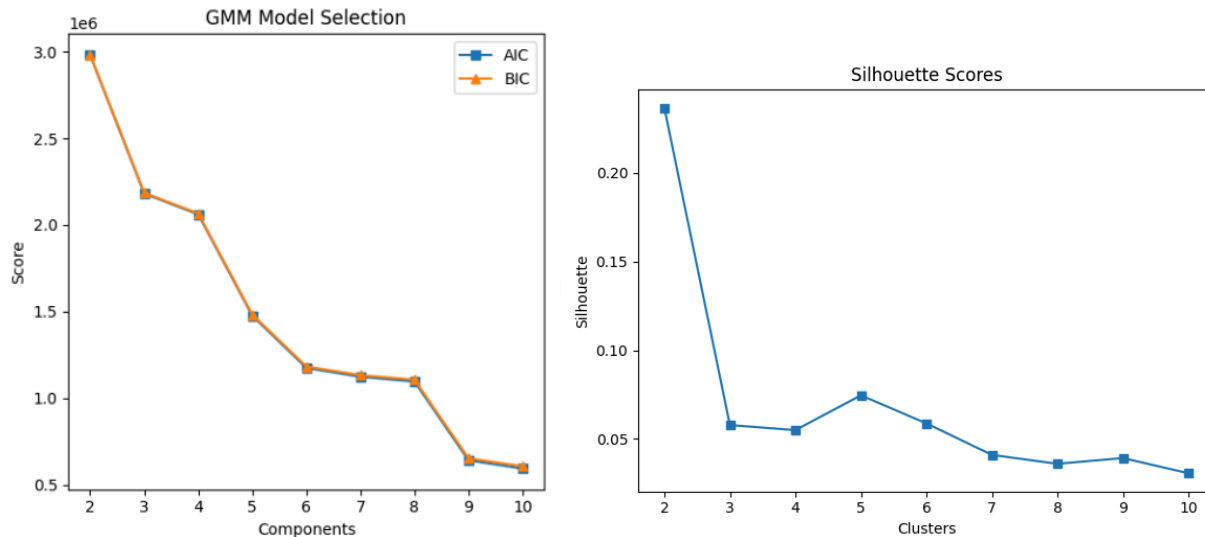
Based on the observed audio features of each cluster, each cluster is assigned their mood as follow:

Cluster	Feature Pattern	Mood
0	High instrumentalness, loud, energetic	Intense
1	Balanced energy/loudness, moderate speechiness	Party
2	Low energy/valence, somewhat loud	Melancholic
3	Very low energy/danceability, high acousticness	Calm

4	Evenly balanced features, moderate tempo	Chill
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GMM Clustering

To determine the optimal number of clusters, the AIC and BIC plots and the silhouette scores are computed.

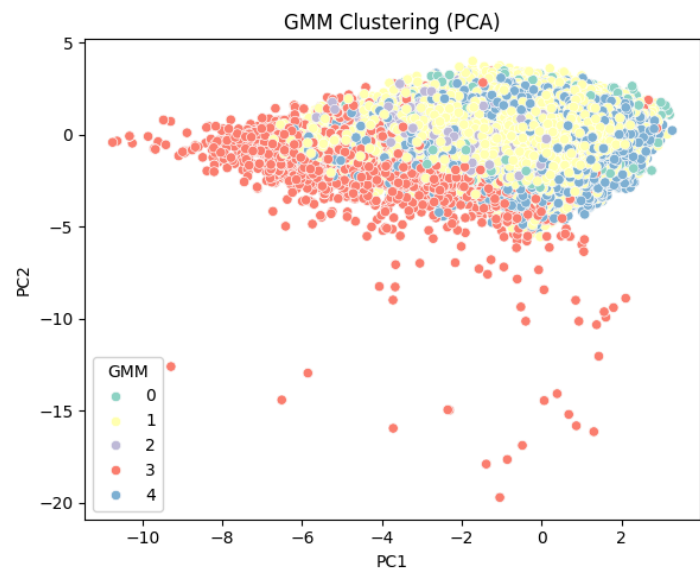


Based on the AIC and BIC curves in the selection graph of the generalized mixture model, we observed a significant sharp decline in the models from 2 components to 5 components, indicating a significant improvement in the model. However, the decrease in AIC and BIC scores began to level off after 5 components, suggesting that the improvement in model fitting effect was no longer significant. On the other hand, when the number of clusters is 2, the silhouette score is the highest, but like the KMeans algorithm, using only two clusters might overly simplify the underlying structure. Among the other options, the performance of 5 components is particularly outstanding, with its silhouette score significantly higher than the adjacent values. Therefore, 5 is selected as the optimal number of components for the GMM model, as it achieves a balance between lower AIC/BIC scores and higher silhouette scores, while avoiding overly rough clustering.

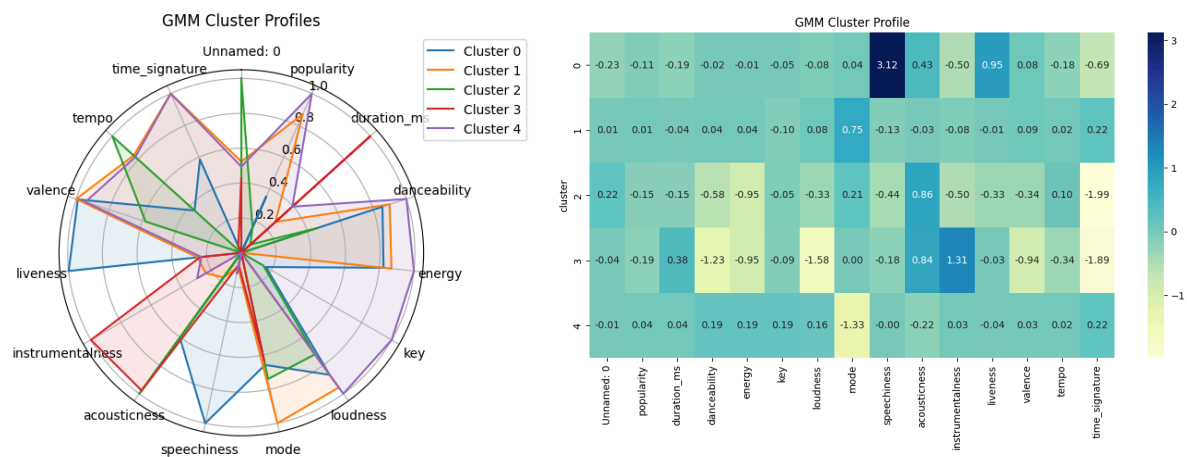
GMM is applied for $k = 5$

```
optimal_k = 5
gmm = GaussianMixture(n_components=optimal_k, random_state=42)
gmm_labels = gmm.fit_predict(X_scaled)
```

The results are visualised using PCA visualisation that reduces the data to 2D.



For more intuitive visualisation, each cluster profile is calculated and visualised.



GMM Interpretation

Based on the observed audio features of each cluster, each cluster is assigned their mood as follow:

Cluster	Feature Pattern	Mood
0	Very high speechiness, high instrumental	Intense
1	High speechiness, balanced energy and danceability	Party
2	Low energy and valence, moderate acoustic	Melancholic
3	Low energy and valence, moderate acoustic	Calm
4	Moderate balance overall, low mode	Chill

Business Insight 2: Clustering Tracks Based on Karaoke Difficulties

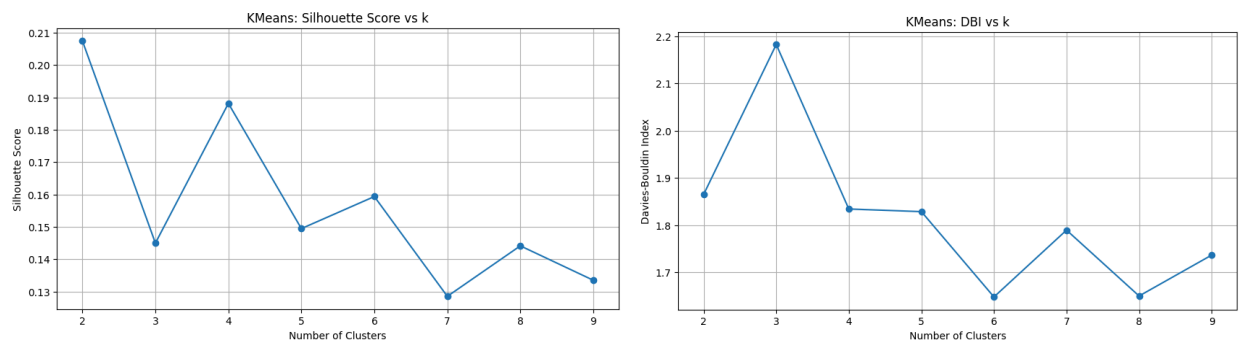
Based on audio features such as energy, key, loudness, mode, speechness, acousticness, instrumentality, liveness, valence and tempo, songs are clustered based on Karaoke difficulties, targeted to help singers choose or challenge pieces that matches their karaoke skill level. This insight can personalize karaoke experiences for users and support targeted practice for vocal improvement.

Data are standardised and columns used are stored in the data frame `df_scaled`.

```
df_scaled = df[['duration_ms_T', 'energy_T', 'key_T', 'loudness_T',  
               'mode_T', 'speechiness_T', 'acousticness_T', 'instrumentality_T',  
               'liveness_T', 'valence_T', 'tempo_T']]
```

K-Means Clustering

To determine the optimal number of clusters for K-means clustering, the silhouette score and DBI score are computed and plotted in a graph.

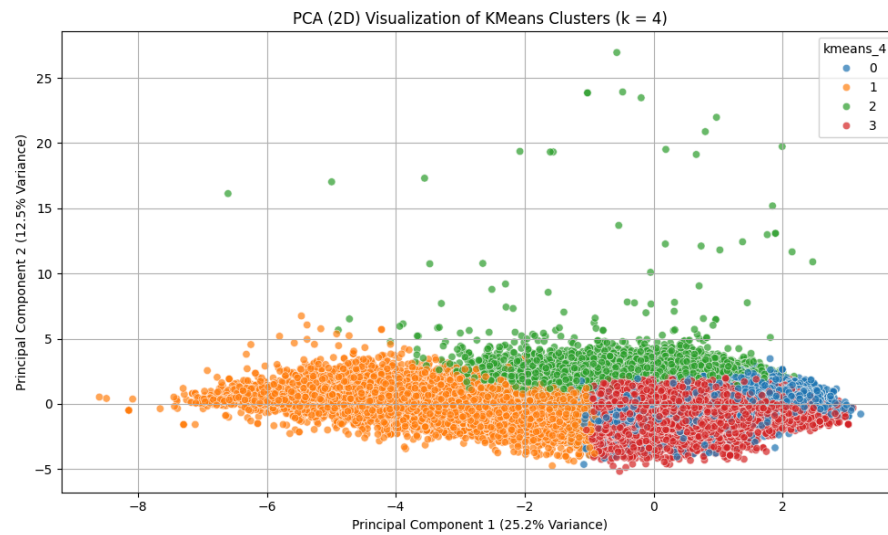


Based on the Silhouette Score plot, the highest value occurs when the number of clusters is 2. However, using only 2 clusters may oversimplify the data and fail to capture meaningful groupings. Although the silhouette score slightly decreases at $k = 4$, it remains relatively high compared to other values. In the Davies-Bouldin Index plot, $k = 4$ shows a noticeable drop from $k = 3$, indicating better separation and compactness of clusters. Therefore, $k = 4$ is selected as the optimal number of clusters.

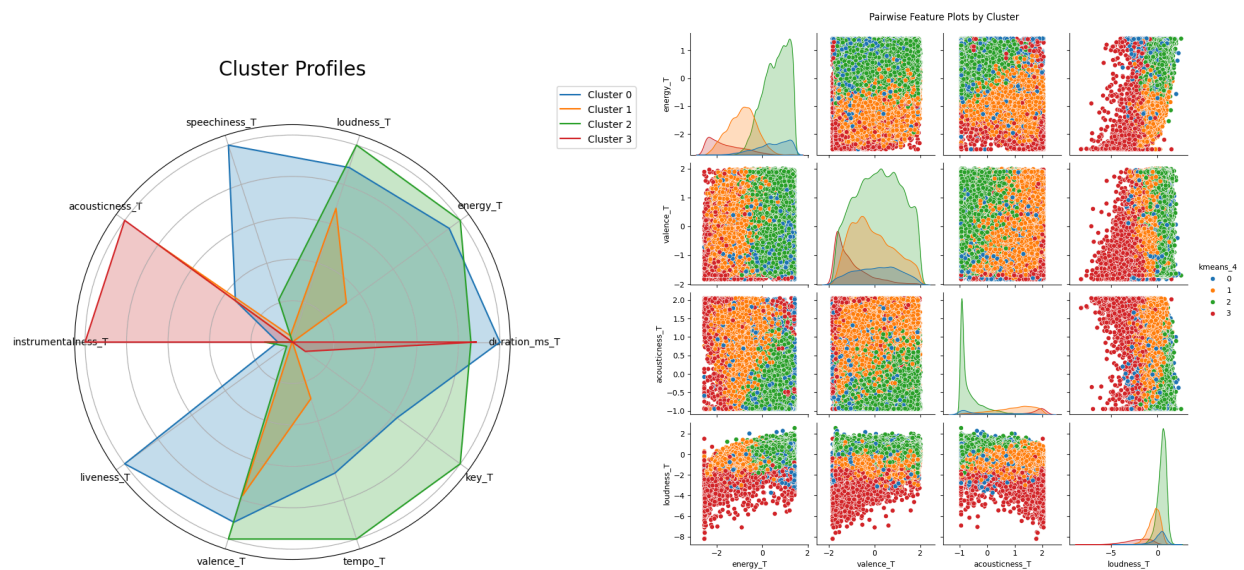
With number of cluster, $k = 4$, KMeans clustering is carried out as follow:

```
kmeans = KMeans(n_clusters=4)  
kmeans.fit(df_scaled)
```

Results are visualised as following:



For more intuitive visualisation, each cluster profile is calculated and visualised.



K-Means Interpretation

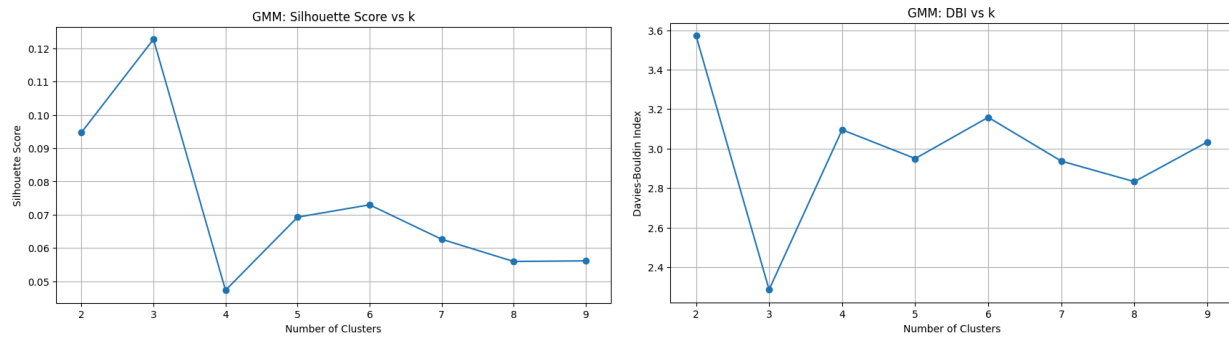
Based on the observed audio features of each cluster, each cluster is assigned their karaoke difficulty as follow:

Cluster	Feature Pattern	Karaoke Difficulty
0	Live tracks, long duration, high speechiness, loudness, energy and valence	Hard
1	Low energy, loudness, tempo and speechiness	Easy

2	Highest tempo, energy, key and valence	Extreme
3	High instrumentalness and acousticness, other metrics are not observable	Not for Karaoke

GMM Clustering

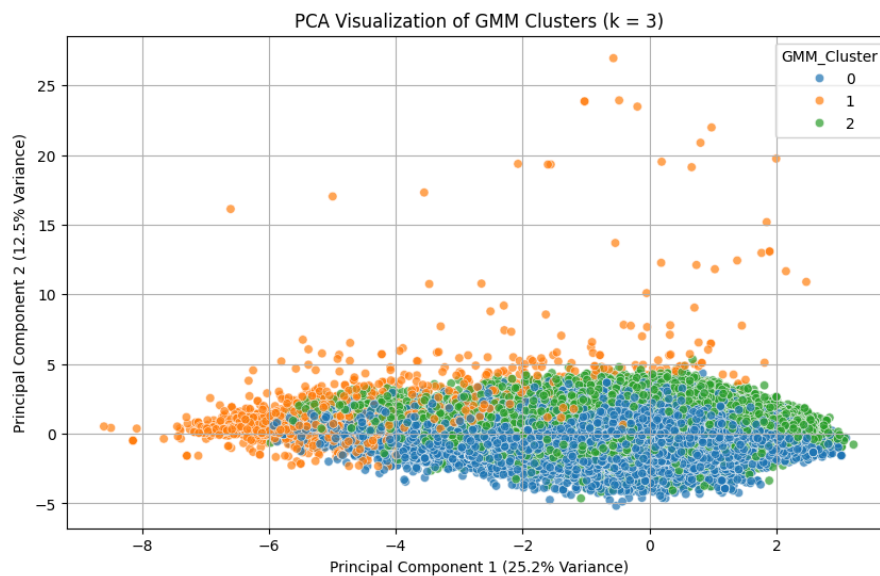
To determine the optimal number of clusters for GMM clustering, the silhouette score and DBI score are computed and plotted in a graph.

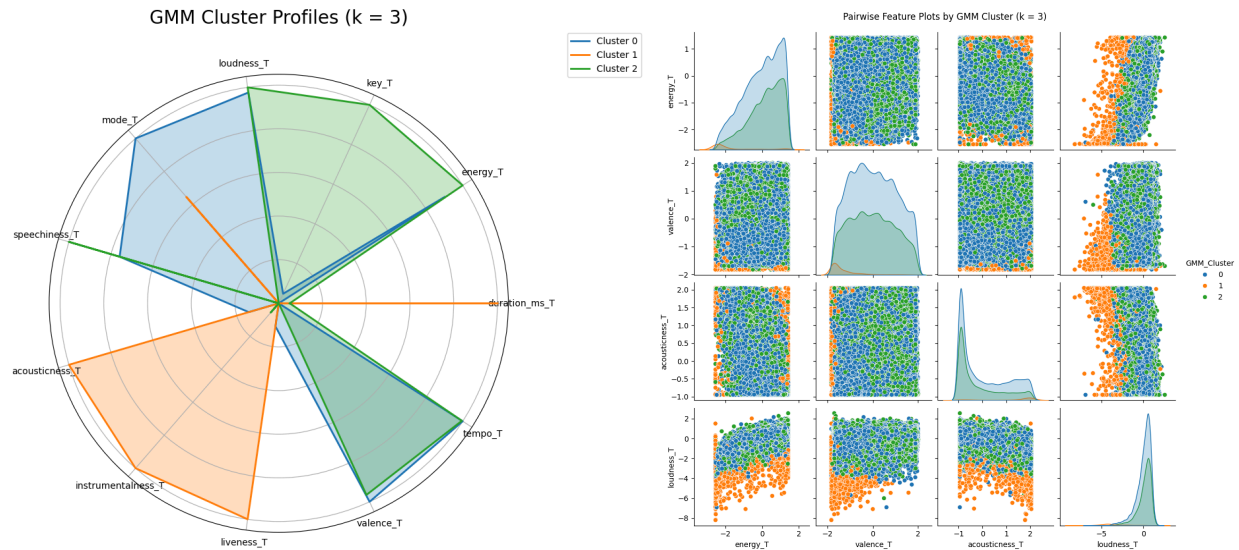


Based on both evaluation metrics, **k = 3** is selected as the optimal number of clusters.

```
gmm = GaussianMixture(n_components=3, random_state=42)
labels = gmm.fit_predict(df_scaled)
```

Results are visualised as following:





GMM Interpretation

Based on the observed audio features of each cluster, each cluster is assigned their karaoke difficulty as follow:

Cluster	Feature Pattern	Karaoke Difficulty
0	High tempo, mode, valence and loudness, noticeable amount of speechiness	Medium
1	Live tracks, high instrumentalness and acousticness	Easy
2	Highest tempo, energy, key, valence and loudness	Hard

Business Insight 3: Clustering Artist

The artists are being clustered according to the features of their respective music.

For this business insight, the data preparation slightly varies from that explained at the beginning of the report. The track's features are first aggregated and grouped by artists.

```
artist_features = df.groupby('artists')[[
    'valence', 'energy', 'danceability', 'tempo', 'acousticness',
    'instrumentalness', 'loudness', 'speechiness', 'liveness'
]].mean().reset_index()
```

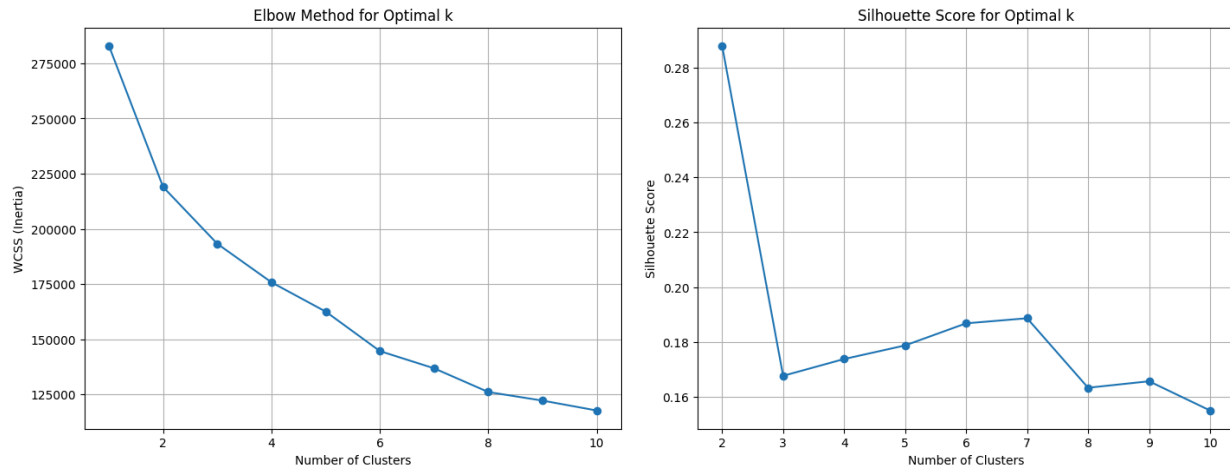
The features are then scaled.

```
scaler = StandardScaler()
```

```
scaled_features =  
scaler.fit_transform(artist_features.drop('artists', axis=1))
```

K-Means Clustering

To determine the optimal number of clusters, the elbow curve and the silhouette scores are computed.

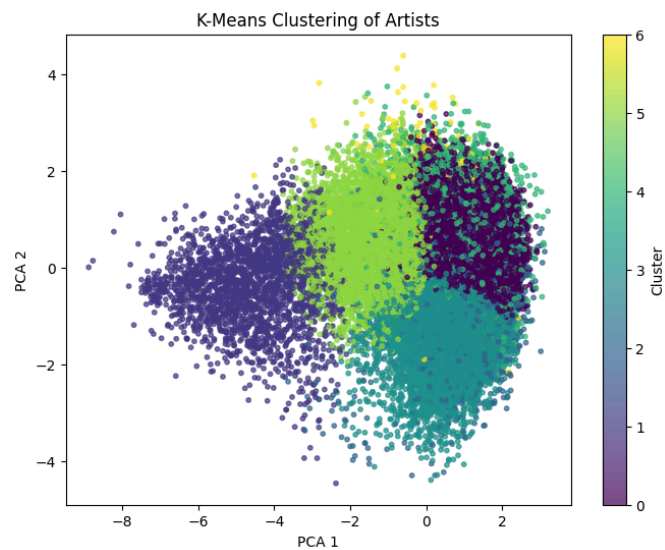


Based on the elbow curve, we can see that there is a steep slope from 1 to 2, but the gradient decreased from 2 to 3. While the silhouette score is the highest when the number of clusters is 2. However, separating the clusters to only two groups may not be meaningful. Instead, 7 is selected as the optimal number of clusters as it has the highest silhouette score aside from 2 clusters. The gradient from 6 to 7 in the elbow curve is also obviously smaller.

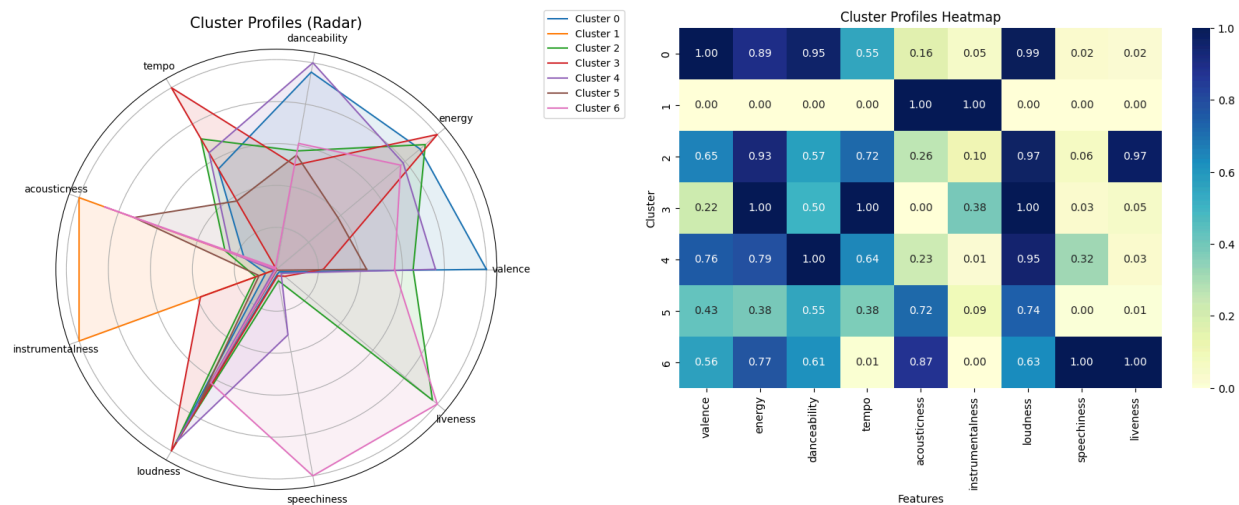
K-means is applied with 7 clusters.

```
kmeans = KMeans(n_clusters=7, random_state=42)  
artist_features['km_cluster'] = kmeans.fit_predict(scaled_features)
```

The results are visualised using PCA visualisation that reduces the data to 2D.



For more intuitive visualisation, each cluster profile is calculated and visualised.



K-Means Interpretation

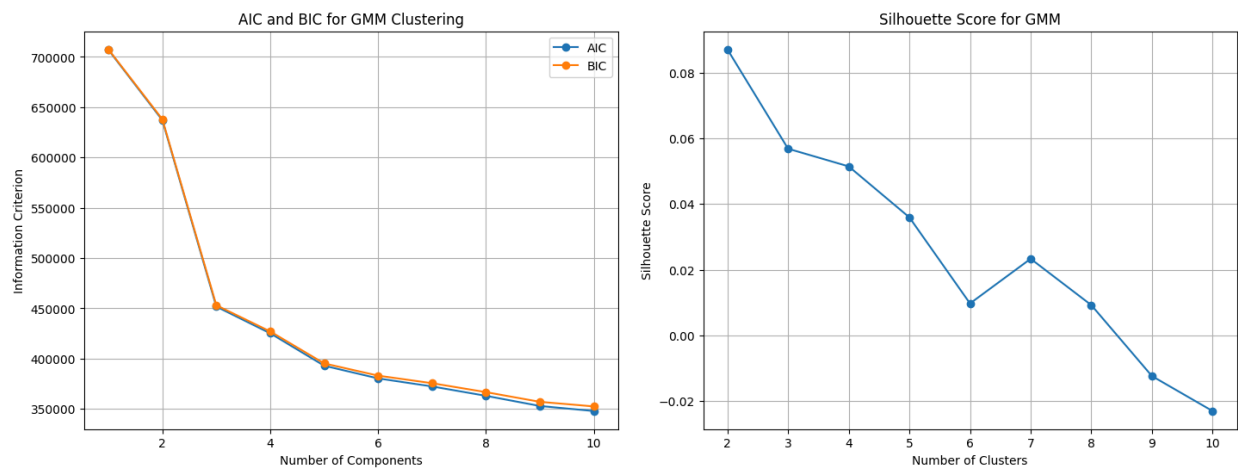
Based on the observed audio features of each cluster, each cluster is assigned their artist style as follow:

Cluster	Feature Pattern	Artist Style
0	High valence, energy, danceability, loudness Low acousticness, speechiness, liveness	Upbeat Dance-Pop Artists
1	High acousticness, instrumentalness Low valence, energy, danceability, tempo, loudness, speechiness, liveness	Calm Acoustic/Instrumental Artists

2	High energy, loudness, liveness Low instrumentalness, speechiness	Live-Performance-Oriented Vocal Artists
3	High energy, tempo, loudness Low valence, acousticness, speechiness, liveness	Aggressive Electronic/Rock Artists
4	High valence, energy, danceability, loudness Low acousticness, instrumentalness, liveness	Mainstream Pop Artists
5	Low instrumentalness, speechiness, liveness	Contemporary Vocal Artists
6	High energy, acousticness, speechiness, liveness Low tempo, instrumentalness	Expressive Acoustic or Live Rap Artists

GMM Clustering

To determine the optimal number of clusters, the AIC and BIC plots and the silhouette scores are computed.

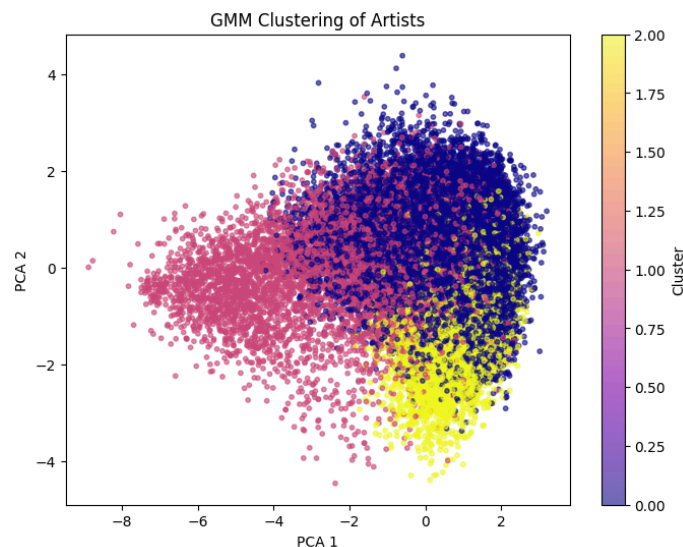


Based on the AIC and BIC, the slope becomes gradual once it reaches 3 numbers of components, and once it reaches $k = 5$, the slope becomes even more shallow. For the silhouette score, 2 clusters would have the highest score but 2 clusters may not be meaningful. Since 3 clusters have a better silhouette score compared to 5 clusters, $k = 3$ is chosen.

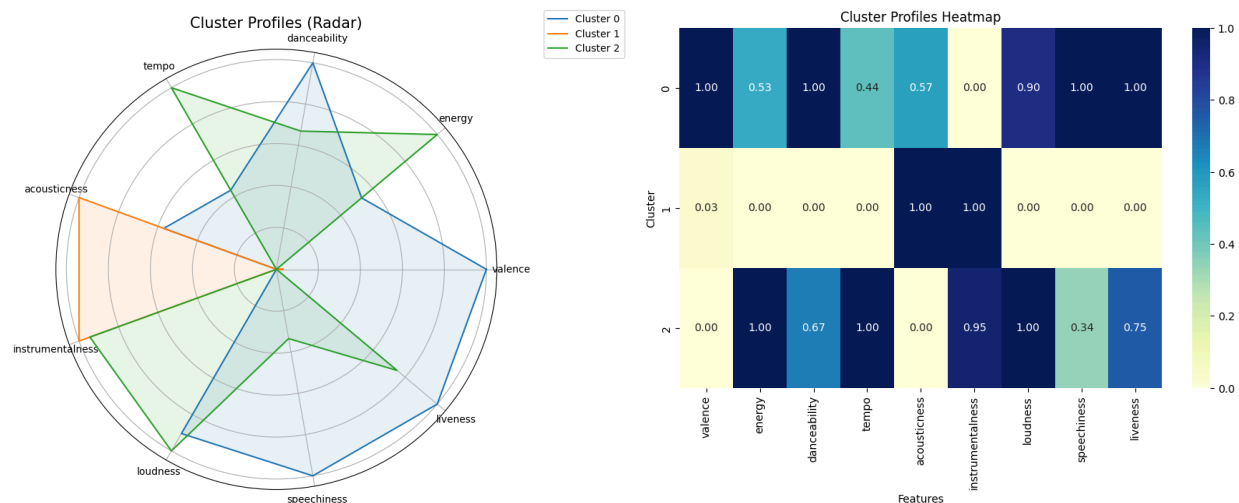
GMM is applied for $k = 3$

```
gmm = GaussianMixture(n_components=3, random_state=42)
artist_features['gmm_cluster'] = gmm.fit_predict(scaled_features)
```

The results are visualised using PCA visualisation that reduces the data to 2D.



For more intuitive visualisation, each cluster profile is calculated and visualised.



GMM Interpretation

Based on the observed audio features of each cluster, each cluster is assigned their artist style as follow:

Cluster	Feature Pattern	Artist Style
0	High valence, danceability, loudness, speechiness, liveness Low instrumentalness	High-energy vocal artists
1	High acousticness, instrumentalness Low valence, energy, danceability, tempo, loudness, speechiness, liveness	Instrumental or ambient-focused artists
2	High energy, tempo, instrumentalness, loudness, liveness Low valence, acousticness	Loud, high-tempo, live-oriented instrumental artists

Model Evaluation

There are three methods to assess the effectiveness of clustering algorithms: external, internal, and relative. External methods cannot be implemented because there are no predefined labels for any of the three business insights identified. The relative method, on the other hand, is used to determine the optimal number of clusters. Therefore, internal methods are employed to evaluate the model.

Silhouette score, Davies-Bouldin Index (DBI), Dunn Index, and Within-Cluster Sum of Squares (WCSS) were used as evaluation metrics, each focusing on different aspects. The silhouette score measures how similar each data point is to its own cluster compared to other clusters, with a higher value indicating better-defined and more distinct clusters. The DBI measures the ratio of within-cluster scatter to between-cluster separation for each cluster, averaging it over all clusters. A lower DBI indicates small within-cluster distances and large between-cluster distances, which is preferable. In contrast to the DBI, which averages over all clusters, the Dunn Index is the ratio of the smallest distance between clusters (separation) to the largest intra-cluster distance (compactness), where a higher value is better. Lastly, the WCSS measures the total squared distance between each point and the centroid of its cluster, with a lower WCSS indicating tighter clusters.

Business Insight 1: Clustering tracks based on mood

The performance of the K-Means Algorithm and GMM are evaluated using the metrics mentioned below.

	Silhouette Score	Davies-Bouldin Index (DBI)	Dunn Index	Within-Cluster Sum of Squares (WCSS)
K-Means (k = 5)	0.0992	2.3826	-	1254254.1146
GMM (k = 5)	0.0746	2.9941	-	1450733.8406

According to the clustering evaluation indicators, the overall performance of the K-Means model with 5 clusters is superior to that of the GMM model with the same number of clusters. The silhouette score of K-Means is higher (0.0992), indicating that the cohesion within the clusters is stronger and the separation between clusters is greater, which is better than that of GMM. A higher silhouette score usually means clear boundaries between groups.

Furthermore, the Davies-Bouldin index of the K-Means algorithm is also lower (2.3826), indicating that the clusters generated by K-Means are more compact and better separated, which is superior to the clusters formed by GMM (whose Davies-Bouldin index is 2.9941).

Due to the differences in the clustering formation logic of different models, WCSS cannot be directly compared between the models. However, it still allows us to get a glimpse of the compactness of the clusters. The K-Means algorithm performs better in this regard, with a WCSS value of 1,254,254.11,

which is lower than that of GMM (1,450,733.84), meaning that each point in a cluster in K-Means is closer to its respective centroid.

The **Dunn Index** calculation involves computing a pairwise distance matrix for all data points. With 114,000 data points, this matrix would be very large (114,000 x 114,000), requiring a significant amount of memory, likely exceeding the available RAM, resulting in the "unable to allocate memory" error.

Overall, K-Means shows more favorable clustering performance, with better-defined and more compact clusters. These results suggest that K-Means is the more effective model for clustering tracks based on mood in this dataset.

Business Insight 2: Clustering Tracks based on Karaoke Difficulties

The performance of the K-Means Algorithm and GMM are evaluated using the metrics mentioned above. Performance of GMM of k = 4 was also recorded for comparison.

	Silhouette Score	Davies-Bouldin Index (DBI)	Dunn Index	Within-Cluster Sum of Squares (WCSS)
K-Means (k = 4)	0.1881	1.8341	0.0009	893853.4
GMM (k = 3)	0.1227	2.2851	0.0057	1016566.0
GMM (k = 4)	0.0473	3.0952	0.0001	975029.0

Based on the clustering evaluation metrics, the K-Means model with 4 clusters demonstrates the strongest overall performance compared to the GMM models with 3 and 4 clusters. The K-Means solution has the highest Silhouette Score (0.1881), indicating better-defined clusters with more separation between them. It also achieves the lowest Davies-Bouldin Index (1.8341), suggesting that the clusters are more compact and well-separated relative to the other methods.

The Dunn Index is slightly higher for GMM with 3 clusters (0.0057) than for K-Means (0.0009), which implies GMM might have a slightly better minimum separation relative to the worst intra-cluster spread. However, Dunn Index alone is less reliable when the other metrics show consistently weaker results. Additionally, the Within-Cluster Sum of Squares (WCSS) for K-Means is the lowest at 893,853.4, meaning its clusters are more tightly packed around their centroids.

Overall, the higher Silhouette Score and lower DBI strongly favor K-Means in this comparison. This suggests that K-Means provides more distinct and compact clusters for your data and is the better choice among the tested models.

Business Insight 3: Clustering Artist

The performance of the K-Means Algorithm and GMM are evaluated using the metrics mentioned above. However, for fairer comparisons, the performance for GMM for 7 clusters is also calculated.

	Silhouette Score	Davies-Bouldin Index (DBI)	Dunn Index	Within-Cluster Sum of Squares (WCSS)
K-Means (k = 7)	0.1887	1.44044	0.0161	136856.94
GMM (k = 3)	0.0568	2.4225	0.0094	23747.11
GMM (k = 7)	0.0233	4.9895	0.0096	201516.40

Based on the evaluation metrics, K-Means (k = 7) outperforms GMM (k = 3) in all aspects except for WCSS. However, WCSS is not directly comparable here, as the number of clusters differs. To ensure a fair comparison, the performance of GMM with k = 7 was also evaluated.

When both models use seven clusters, K-Means clearly outperforms GMM across all internal evaluation metrics. The higher Silhouette Score shows that K-Means clusters are more distinct and less overlapping. The lower Davies-Bouldin Index (DBI) suggests that K-Means clusters are more compact and the higher Dunn Index reflects better worst-case separation and intra-cluster tightness. Besides, GMM shows a higher WCSS, indicating a less optimised solution with more overlapping components.

In short, K-Means provided more distinct, compact and well-separated clusters than GMM, as reflected by all internal validation metrics.

Conclusion

Across all three business insights, the K-Means algorithm consistently outperformed the GMM algorithm. It is preferred due to its ability to produce tight, well-separated clusters, as indicated by the evaluation metrics. This could be a result of the dataset's characteristics, as K-Means tends to perform well when the clusters are roughly spherical or of similar sizes. In contrast, GMM assumes that the data follows a mixture of Gaussian distributions, which may not be an appropriate assumption for this dataset. Therefore, K-Means appears to be the more effective algorithm for clustering in this particular case.

Reflection

Lam Yoke Yu

Part A: Project Reflection

In Project 1, we were required to perform classification tasks and were advised to use data mining tools. I downloaded KNIME for this purpose; however, I had difficulty understanding how to use the tool. In the end, my group members and I decided to use Python Notebooks to complete the task. Although it was suggested that each team member use one algorithm, we decided to divide the work by business insight instead, as the preprocessing and feature selection for each insight would be the same across all classification algorithms. This allowed us to simply plug in the training set into the respective function.

The business insight I was responsible for was the first one, Predicting Hit Songs Using Popularity Correlations. I remember that initially, my SVM model produced all zero values for precision, recall, and F1-score, as it classified everything as negative. At first, I thought this was acceptable, providing a justification for the result. However, during our cross-checking on the day of submission, a team member reminded me that such a result was not acceptable. After discussing it with the lecturer, I realized the result was indeed problematic. I had used the `LinearSVC()` model instead of `SVC()`, while my data was not linear. I felt quite anxious as the model took over 20 minutes to run, while the other models only took a few minutes. I restarted the kernel several times, but eventually decided to leave it running. The result came out after 48 minutes of runtime. For Project 2, we continued the task distribution method from Project 1, and I was assigned to work on the third insight which is clustering artists.

I find the entire project process to be somewhat magical. It felt like we simply cleaned the data, selected certain attributes, fed them into the model, and boom, the results appeared.

Part B: Course Reflection

Throughout the course, I learned about the data mining process and various machine learning techniques. It was really interesting to explore methods such as classification, clustering, and association. This course helped me understand the basics of how algorithms work and how different algorithms are suited to different types of data and tasks. However, I am still intrigued by how these algorithms work beyond simple examples, and fascinated by the complexities behind machine learning models, specifically, how they handle large datasets, what makes one model more suitable than another in specific situations, and how trained models are deployed in real-world applications. I plan to study these algorithms in more depth during my upcoming semester break.

Tan Yi Ya

Part A: Project Reflection

In this project, we are using the same dataset that we chose in the first phase, which is the Spotify music tracks dataset from Kaggle. I did not regret choosing this dataset because it has a lot of continuous data which is suitable for our task. In the first phase, we were asked to draft six business insights from the dataset, and I was so afraid to do it wrong. Because we had no prior knowledge or experience with data mining or machine learning, I read out lecture slides, searched on websites and YouTube, got suggestions

from AI, and after the hands-on tasks we did (classification and clustering), I realized I still got my business insights wrong at that time. That's why I want to emphasize that project-based learning is the best way of really diving into the topic. I was thankful to have been given this project and my teammates who worked alongside me. In the future, I want to explore other ways of machine learning or visualization tools aside from using Python.

Part B: Course Reflection

In this course, there are five topics which introduce us to what data mining is, data preprocessing, classification, clustering, and association rule mining. Our lecturer would give us tasks to explore each of the topics in groups under her guidance and turn them into slides or notes to share with our classmates. The topic that made the biggest impression on me was the one I presented in front of everyone, which was the genetic algorithm in classification. Aside from that, we were given a group project in groups of three, carried out in sequential order, from identifying business insights to doing classification and clustering on the dataset we chose. Overall, I learned a lot from the data mining course and understand that raw data is not useful if we don't extract insights from it. In the future, I would like to apply the knowledge I learned not only for work, but also in daily life, such as exploring model building to predict stock market trends.

Tan Zhi Ming

Part A: Project Reflection

For our project, my team decided to use the same dataset for both Project 1 (Classification) and Project 2 (Clustering). In Project 1, which involved a classification task, I initially aimed to predict the type of track based on features such as energy and acousticness. However, I encountered a significant challenge with overfitting. Despite several attempts to resolve the issue—such as adding more features—the model still performed poorly. As a result, I shifted my business insight to a new angle: Classifying Songs by Instrumentation Level Using Audio Features. Although I had to change direction, the experience taught me a valuable lesson about overfitting, a common issue in real-world machine learning applications. It gave me a deeper understanding of model generalization and the importance of proper feature selection.

For Project 2, which focused on clustering, I faced difficulty identifying the optimal number of clusters (k). Fortunately, with the guidance of my teammates, I was able to work through the process step by step using appropriate evaluation metrics. Another challenge I encountered was when I tried to compute the Dunn Index metric—my system ran out of memory, and I was unable to complete the calculation. This experience made me more aware of the hardware requirements needed for processing large datasets and the limitations of my current setup.

Overall, this project has significantly strengthened my foundational skills in data mining. I gained hands-on experience in handling real challenges like overfitting, feature engineering, model evaluation, and memory management—skills that are crucial in both academic and professional data science work.

Part B: Course Reflection

Through this course, I have learned a wide range of algorithms used in data mining and gained hands-on experience with essential tools such as Jupyter Notebook. One of the most valuable lessons was understanding how to process data from the very beginning—starting with data cleaning, exploration, and

transformation—before applying any model. This course also helped me realize just how valuable data truly is, and how meaningful insights can be extracted when it's properly analyzed. The practical assignments and projects allowed me to apply theory to real-world scenarios, deepening my understanding of both the technical and strategic aspects of data mining.

Overall, I believe the knowledge and skills I've gained from this course serve as a strong milestone in my journey toward becoming a data engineer. It has built a solid foundation that I can confidently carry forward into future studies and professional work.